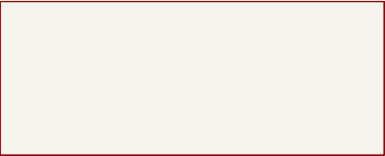


Ghost Fader

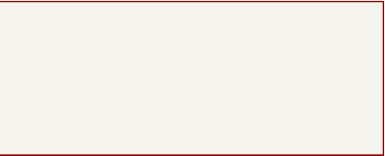
USB-MIDI-controlled volume and pan for two balanced 1/4" line channels. Bus powered, no external supply.
Sheets: 1 input, 2 matrix, 3 output, 4 control, 5 power. Nets between sheets are global labels; rails are power symbols.
Two grounds: GND_USB (XIAO, USB) and GND (audio). Only PS1 (power) and U8 (control) cross between them, plus C45 / R17.

Balanced inputs



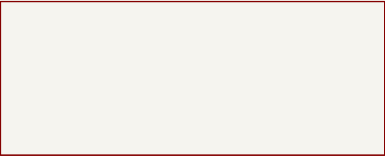
File: input.kicad_sch

2x2 gain matrix and summers



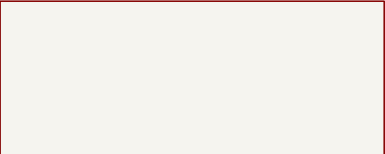
File: matrix.kicad_sch

Balanced outputs



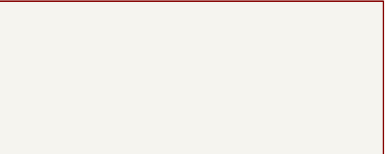
File: output.kicad_sch

Control



File: control.kicad_sch

Power



File: power.kicad_sch

Signal flow: J1/J2 -> U1/U2 THAT1246 (-6 dB) -> C3-C6 -> U3/U4 PGA2310 2x2 matrix -> R2-R9 + U5 OPA2134 summers (x2) -> U6/U7 THAT1646 (+6 dB) -> J3/J4
Control: XIAO RP2040 USB MIDI -> U8 IS07762F -> SPI (SCLK, SDI, /CS) + /MUTE -> U3 -> U4. Plug sense from J1/J2 ring-normal contacts, back through U8.
Power: XIAO 5V pin (USB 5 V) -> FB1 -> PS1 TBA 2-0522 -> +/-12 V (audio ground). +12 V -> U10 78L05 -> +5VA for PGA logic and U8 side 2.

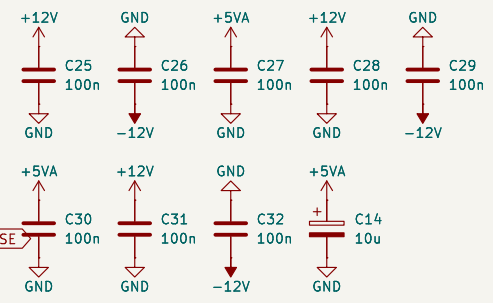
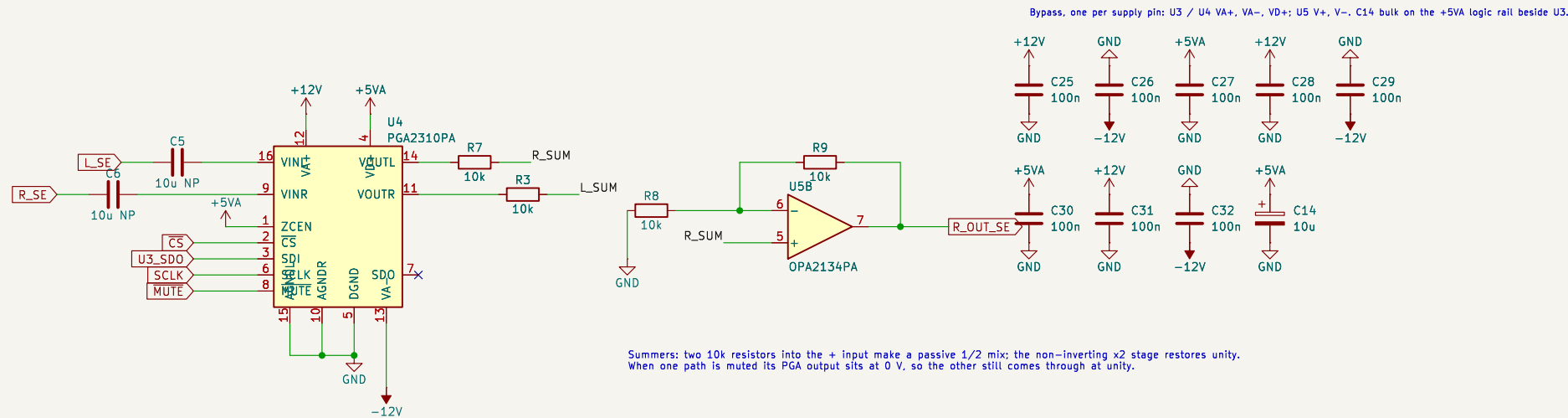
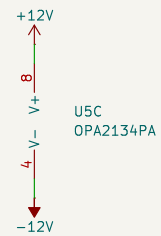
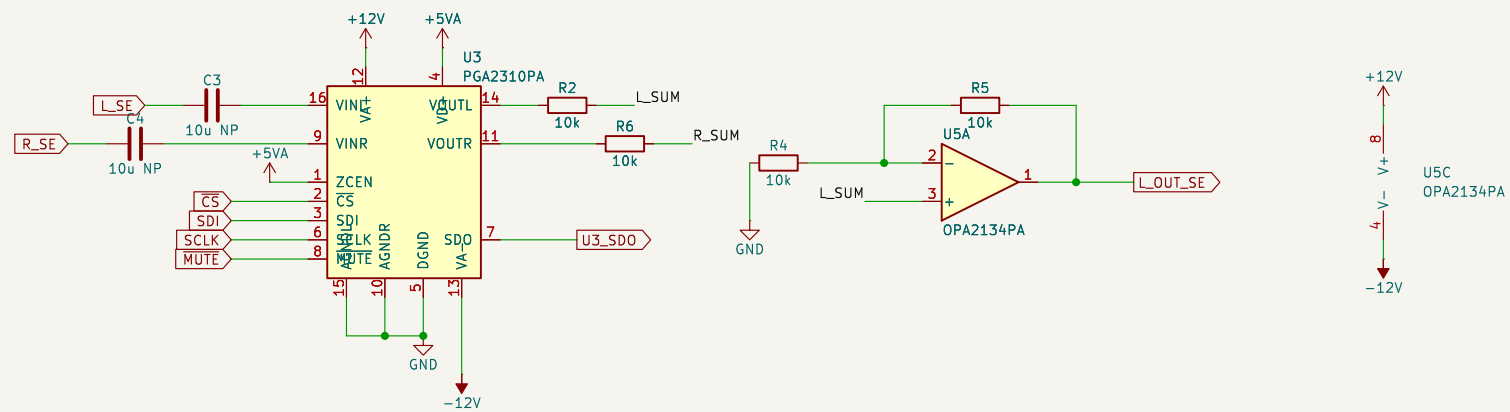
<https://chris-albert.github.io/ghost-fader/>
USB-MIDI volume and pan for two balanced line channels
Ghost Fader

Sheet: /
File: ghost-fader.kicad_sch

Title: Ghost Fader – Rev D

Size: A4	Date: 2026-09-09	Rev: D
KiCad E.D.A. 9.0.6		Id: 1/6

Size: A3	Date: 2026-09-09	Rev: D
KiCad E.D.A. 9.0.6		Id: 2/6



Summers: two 10k resistors into the + input make a passive 1/2 mix; the non-inverting x2 stage restores unity. When one path is muted its PGA output sits at 0 V, so the other still comes through at unity.

U3 = direct paths (L->L on its left channel, R->R on its right). U4 = cross paths (L->R on its left channel, R->L on its right). SPI chain: XIAO MOSI (D10) -> U8 isolator -> U3 SDI, U3 SDO -> U4 SDI. 32 clocks per update; U4 word first, then U3; each word is right byte then left byte, MSB first. ZCEN high: gain changes wait for a zero crossing. Code 192 = 0 dB. 0 = mute. Logic supply is +5VA from U10 (sheet 5), on the audio ground; SDO stays on this side.

<https://chris-albert.github.io/ghost-fader/>
PGA2310 pair, OPA2134 summers

Ghost Fader

Sheet: /2x2 gain matrix and summers/
File: matrix.kicad_sch

Title: Ghost Fader – Sheet 2: 2x2 gain matrix and summers

Size: A3	Date: 2026-09-09	Rev: D
KiCad E.D.A. 9.0.6		Id: 3/6

A



A



D

E

3

1

